

Problem Sheet 6

1. Define $\{\mathbf{Y}_t\}$ as follows:

$$\begin{cases} Y_{1,t} = e_{1,t} \\ Y_{2,t} = e_{2,t} + (-1)^t e_{1,t} \end{cases}$$

where $e_{1,t}$ et $e_{2,t}$ are two independent white noise processes.

- (a) Examine the stationarity of the processes $Y_{1,t}$ and $Y_{2,t}$.
 - (b) Are $Y_{1,t}$ and $Y_{2,t}$ jointly stationary ?
2. The Canadian quarterly time series (1980-2000) for labour productivity (prod), employment (e), unemployment rate (U) and real wages (rw), are in the **Canada** object in the **vars** package. To load and print the dataset in R, use:

```
library(vars)
data(Canada)
print(Canada)
```

- (a) Plot the time series. Describe all systematic patterns you see in the plot.
- (b) Check the stationarity of each component of the multivariate time series.

```
adf_test_Canada <- lapply(Canada, adf.test)
```

- (c) Fit a VAR(1) model to the data. Write out the equation of the model.

```
model <- VAR(Canada, p = 1, type = "both")
summary(model)
res<- residuals(model)
library(MTS)
mq(res)          # Perform a multivariate Ljung-Box test
```

- (d) Does prod (productivity) Granger-cause e (employment)? Does employment (e) Granger-cause productivity (prod)?
- (e) Forecast the next 12 values of the series, and list the forecasted values. Also plot the series along with the forecasted values and 95% prediction limits for the next 12 values of the series.

```
forecast <- predict(model, n.ahead = 12)      # Forecast using the VAR model
fanchart(forecast, main = "VAR Forecast")     # Plot the forecasts
```

3. Consider the dataset which consists of daily closing prices for five major indices: the DAX (Germany), SMI (Switzerland), CAC (France), FTSE (UK), and the S&P 500 (USA).

- (a) Execute the following R code to load, align, and synchronize the multivariate time series:

```
library(vars)
library(quantmod)
library(qgraph)
library(tseries)
library(xts)

# Load European data and Fetch S&P 500
data(EuStockMarkets)
```

```
getSymbols("^GSPC", from="1991-01-01", to="1998-12-31", auto.assign = TRUE)
SP500_prices <- Cl(GSPC)
```

```
# Create a proper date index for European data
eu_dates <- seq(as.Date("1991-07-01"), length.out = nrow(EuStockMarkets)*2, by = "1 day")
eu_dates <- eu_dates[!format(eu_dates, "%u") %in% c("6", "7")][1:nrow(EuStockMarkets)]
EU_xts <- xts(coredata(EuStockMarkets), order.by = eu_dates)
```

```
# Combine into a 5-component system
Combined_Data <- na.omit(merge(EU_xts, SP500_prices, all=FALSE))
colnames(Combined_Data) <- c("DAX", "SMI", "CAC", "FTSE", "SP500")
```

- (b) Plot the five price series. Describe all systematic patterns you observe. Do these series appear to be stationary?

```
plot(Combined_Data, main = "Global Stock Indices")
```

- (c) For financial time series, we often work with returns rather than price levels to satisfy stationarity assumptions. Transform the price data into log-returns using the formula $r_t = \log(P_t) - \log(P_{t-1})$.

```
Returns <- diff(log(as.matrix(coredata(Combined_Data))))[-1,]
```

- (d) Fit a Vector Autoregressive model of order $p = 2$ to the returns. Write out the algebraic equation for the model.

```
model <- VAR>Returns, p = 2, type = "const")
summary(model)
```

- (e) Estimate the Partial Correlation Graph using the `qgraph` package.

```
cor_matrix <- cor>Returns)
graph_model <- EBICglasso(cor_matrix, n = nrow>Returns))
qgraph(graph_model, layout = "spring", labels = colnames>Returns),
       groups = list("Europe" = 1:4, "US" = 5), color = c("lightblue", "orange"))
```

Identify two indices that have a high correlation but no direct edge between them. How do you explain this?